

29/Sept

Full stack development test

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Btech CSE 'B' / CU24250227 / 20.

Q1. `console.log('...');`

```
set timeout() => {  
  console.log('...');  
};
```

```
Promise.resolve().then(() => {  
  console.log('...');  
}));  
console.log('...');
```

Synchronous code → Promise microtasks → timer callback

Explanation:

- Normal console log execute first.
- `Promise.then()` runs after synchronous code finishes
- `set timeout()` runs later in the timer phase.

Q2. two functions, `add(a,b)` & `subtract(a,b)`

calc.js structure:

```
function add(a,b) {  
  }  
function subtract(a,b) {  
  }  
module.exports = {  
  add,  
  subtract  
};
```

~~app.js~~ app.js

```
const {add, subtract} = require('./calc');  
console.log(add(1,1));
```

calc.js → exports functions

↓
app.js require() imports function

Ques 3. Middle ware & Routing

You need an express app with:

- Require Auth middle ware.
- checking the x-api-key header
- unauthorized - request → HTTP 401
- get /api/books public
- post /api/books protected

```
const express = require('express');
```

```
const app = express();
```

```
app.use(express.json());
```

```
function requireAuth(req, res, next) {
```

```
  const apiKey = req.header['x-api-key'];
```

```
  if (apiKey !== '...') {
```

```
    return res.status(401).json({
```

```
      error: "unauthorized"
```

```
    });
```

```
  };
```

```
  next();
```

```
}
```

```
app.get('/api/books', (req, res) => {
```

```
});
```

```
app.post('/api/books', requireAuth, (req, res) => {
```

```
});
```

```
app.listen(5000);
```

Q4. Building a Rest api.

```
const express = require('express');
```

```
const app = express();
```

```
app.use(express.json());
```

```
let books = [];
```

```

app.get('/api/books', (req, res) => {
  });
app.post('/api/book', (req, res) => {
  });
app.delete('/api/books/:id', (req, res) => {
  });
app.listen(5000);

```

Ques 5. Debugging & Error Handling

```

app.get('/api/books/:id', (req, res) => {
  const book = books.find(
    b => b.id == req.params.id
  );
  if (book) {
    return res.status(404).json({
      error: "Book not found"
    });
  }
  res.json(book.title);
});

```

code searches → book title.
 The problem is that if
 find() doesn't find anything
 book is undefined therefore
 accessing title cause an error.

Ques 6 API consuming a Rest API.

Import { use effect, useState } from react;

function Book List() {

```

  const [books, setBooks] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState("");

```

useEffect(() => {

fetch('http://localhost:5000/api/books')

• then(response => {
 })

• then(data => {
 })

• catch(err => {
 })

• finally(() => {
 });

});

if (loading) {

return <p>loading ... </p>;

}

if (error) {

return <p>{error}</p>;

}

return

<div>

{/* display book */ </div>};

export default BookList;

Ques 7. APIs - CORS.

const express = require('express');

const cors = require('cors');

const app = express();

app.use(cors({
 origin: 'http://localhost:3000' }));

app.use(express.json());

app.listen(5000);

The browser treats these as different origin needs to allow the React origin. Postman doesn't enforce browser CORS rules in the same way.